

Chapter 112 — Nocturia

- **Nocturia is voiding during the hours of intended sleep.** It carries substantial morbidity: a **risk factor for falls and fractures**, a **major disruptor of normal sleep architecture** and a potential **harbinger of serious systemic disease**.
- **Aetiology is multifactorial — "Sleep CALM":** Sleep disorders, Comorbidities, Actions (modifiable behaviours and lifestyle factors), Lower urinary tract dysfunction, Medications.
- **One common mechanism:** every Sleep CALM factor disturbs the balance between **nocturnal urine production, storage capacity and normal sleep architecture** — and that balance is best characterised by the **bladder diary**.
- **Evaluation:** systematic yet individualised — **history and examination** → **tailored ancillary testing**.
- **Management:** **optimise the Sleep CALM factors first**; pharmacotherapy targeting **excess nocturnal urine production** is reserved for appropriately selected patients refractory to conservative management.

GUIDELINE SOURCES FOR NOCTURIA

Body	Document	Nocturia content
American Urological Association (AUA)	BPH/LUTS guideline (2021); AUA/SUFU OAB guideline (2019)	No specific nocturia guideline ; insights only by extension from benign prostatic hyperplasia (BPH)/lower urinary tract symptoms (LUTS) and overactive bladder (OAB) guidance (SUFU: Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction)
European Association of Urology (EAU)	Non-neurogenic Male LUTS and Female LUTS guidelines (2022)	Each contains a dedicated nocturia section
EAU Male LUTS guidelines panel	Standalone nocturia guideline (2017)	Detailed evaluation and management guidance
International Continence Society (ICS)	Nocturia consensus statement (2019); nocturia terminology report (2019)	Detailed guidance; the terminology report supplies the definitions used throughout

- **Pattern:** explicit nocturia guidance is **EAU and ICS**; the AUA addresses it only indirectly.

NOCTURIA—WHAT, WHEN, AND WHY?

- **ICS definition (2019):** the **number of times urine is passed during the main sleep period** — from the time a person intends to sleep until the time they intend to rise.
- **The main sleep period is defined by the individual's own sleep schedule**, not by societal norms or the solar cycle (sunset to sunrise). "Night-time" is used for convenience but technically means the individual's sleep period.
- **Shift workers:** nocturia is pervasive across all work and sleep schedules, and similar mechanisms operate **independent of the solar cycle**.
- Current terminology **does not exclude patients who void only once per night** and **does not incorporate bother**.

Why the patient gets up: the definitional shift

ICS 2002 position statement (landmark)	ICS 2019 terminology (current)
"The complaint that the individual has to wake at night one or more times to void "	Number of times urine is passed during the main sleep period — no required cause
"[In order] to void" implied that urge-driven awakenings were distinct from waking for poor sleep or other non-urological reasons	Omission of the specifier recognises that urge-driven awakening and poor sleep are intertwined and the aetiology multifactorial

- **Sleep-related and other "non-urological" factors are not grounds for excluding a patient from nocturia treatment; they are leverage for an interdisciplinary approach.**

From the guidelines

- **AUA BPH/LUTS:** nocturia is **often multifactorial and symptomatic of other medical problems** — a unique symptom complex requiring judicious evaluation.
- **AUA/SUFU OAB:** nocturia is often due to factors unrelated to OAB; **isolated nocturia is a separate symptom entity** needing different evaluation and management.
- **ICS consensus:** nocturia was long treated as merely one LUTS of OAB or benign prostatic obstruction; it is now recognised as a **specific symptom in its own right** with wide-ranging pathophysiology.

KEY POINTS: NOCTURIA—WHAT, WHEN, AND WHY?

- **Voids during the main period of intended sleep**, set by the patient's own pattern, not the time of day.
- **Multifactorial**; may arise for reasons other than urge.

TABLE 112.1 KEY TERMS AND DEFINITIONS

Term	Definition / formula	Interpretation
Nocturia severity and sleep		
Main sleep period	Going to bed intending to sleep → waking intending to rise	Individual sleep cycle, not solar cycle
Nocturia	Voids during the main sleep period; each preceded and followed by sleep or the intention to sleep	—
Night-time frequency / actual number of nightly voids (ANV)	Voids recorded within the main sleep period	The severity count
First uninterrupted sleep period (FUSP) (h)	Start of main sleep period → first nocturnal void	Key sleep measure (below)
First morning void	First void after waking with intention to rise	Counts as daytime frequency
Nocturnal enuresis	Incontinence during the main sleep period	—
Urine production		
Total urine volume (TUV) (mL)	Urine produced over 24 h	—
Nocturnal urine volume (NUV) (mL)	All voids during the night + first morning void	Volume produced during the main sleep period
Nocturnal polyuria index (NPI)	NPI = NUV / TUV	Proportion of 24-h output produced during sleep

Term	Definition / formula	Interpretation
Nocturnal urine production (NUP) (mL/h)	NUP = NUV / duration of main sleep period	Hourly production rate during sleep
Bladder storage capacity		
Maximum voided volume (MVV) (mL)	Largest single void in 24 h	Proxy for maximum functional capacity
Nocturnal maximum voided volume (NMVV) (mL)	Largest single void during the main sleep period	Proxy for nocturnal functional capacity
Production-storage mismatch		
Nocturia index (Ni)	Ni = NUV / MVV	>1: NUV exceeds maximum capacity → nocturia or enuresis
Modified nocturia index (MNI)	MNI = NUV / NMVV	>1: NUV exceeds nocturnal functional capacity
Predicted number of nightly voids (PNV)	PNV = Ni – 1	Used to calculate NBCi
Nocturnal bladder capacity index (NBCi)	NBCi = ANV – PNV	>0: voids occur below maximum capacity; >2: severe nocturia
Diary-based mechanisms		
Nocturnal polyuria (NP)	Excess urine production during the main sleep period	Commonest: NPi >20–33% (age-dependent); NUP >1.5 mL/min (>90 mL/h) . Others: NUV >6.4 mL/kg; nocturnal production >0.9 mL/min (54 mL/h)
Global polyuria	Excess production across 24 h	>40 mL/kg/24 h (>2.8 L at 70 kg)
Diminished bladder capacity	Capacity expected to cause nocturnal symptoms at normal urine volume; nocturnal-specific or global	No consensus. Normal capacity usually >300–400 mL ; MVV or NMVV <150–300 mL suggestive

Terms modified from Hashim et al., ICS report on the terminology for nocturia and nocturnal lower urinary tract function, 2019.

- **What the table is for:** it converts a diary into three quantities — **how much is made (NUV, NPi, NUP), how much can be held (MVV, NMVV), and the mismatch between them (Ni, NBCi).**
- **Pattern:** every mismatch index is **volume ÷ capacity**; the "modified" index uses NMVV, so needs night data only.
- **Trap — the first morning void is split:** its volume counts towards NUV, but the void counts as daytime frequency, not as a nocturia episode.
- **Trap:** global polyuria is **weight-indexed**; NP is a **proportion or a rate** — different bases, not mutually exclusive.

Concept — Nocturia is defined by the patient's **intent and own sleep schedule**, not the clock or the cause. Every void within intended sleep counts, and the question of why it happened is deliberately left open for a multifactorial work-up.

RATIONALE FOR EVALUATION AND MANAGEMENT

Nocturia Is Common

- **Older age is among the most robust correlates**, but nocturia is common across ages, races and sexes worldwide.

- **1 in 5–6 people aged 20–40 years void ≥ 2 times per night; up to 3 in 5 people >70 years void ≥ 2 times nightly.**
- **Boston Area Community Health (BACH) survey (landmark): higher prevalence in Black and Hispanic than White respondents; partly mediated by socio-economic status** — a disparity in urological practice requiring investigation.

Nocturia Is Bothersome

- Among the **most bothersome urinary complaints** in the general population; bother is linked to **reduced quality of life (QoL)** and **more depressive symptoms**.
- **High bother** → more difficulty **initiating sleep, returning to sleep** after a void, and more **morning fatigue**.
- Most people with **≥ 2 voids per night** report significant bother; but frequency and bother correlate only at population level — **one void per night can be highly bothersome**.
- **Bother drives help-seeking**; many with nocturia never seek help. Nocturia may **dominate the interview and mask other LUTS**.
- **Any nocturia that brings a patient to urological care is, by definition, clinically significant**, irrespective of the number of voids.

Impact of Nocturia

Sleep

- Harm is mediated largely through **sleep architecture**; **urge-driven awakening is the commonest cause of adult sleep disruption**.
- Affects **sleep latency** (time to fall asleep after lights out), **sleep maintenance** and **sleep efficiency** (percentage of time in bed spent asleep).
- An **independent predictor of insomnia and reduced sleep quality in older adults**; worsens wakefulness in baseline insomnia. **Polysomnography** confirms an independent association with poor sleep **across the whole night**.
- **Harm far exceeds the minutes lost to toileting** — because of timing.

SLEEP ARCHITECTURE — THE TIMING PROBLEM

Early night	Late night
Slow-wave sleep (N3) is densely clustered here	Rapid eye movement (REM) sleep predominates
Deepest, most restorative stage; disruption is not recovered later in the night	Interruption here is less costly to restoration
Most first voids occur within 2–3 h of sleep onset → land in slow-wave sleep	—

- **Structure:** non-REM (NREM) $\approx 75\%$ of sleep, stages **N1** (transition), **N2** (deeper), **N3** ("slow-wave", deepest); cycles alternate NREM and REM, **4–6 cycles per night**; slow-wave sleep is only **10–25% of total sleep**.
- **Consequence:** daytime fatigue and poor sleep quality **even when total sleep time is preserved**.
- Home electroencephalography: **less slow-wave sleep when the first void falls in the first two sleep cycles**. A **shorter interval to first void** → worse whole-night sleep quality and more daytime dysfunction.

Falls and fractures

- Nocturia is among the **LUTS most strongly associated with falls and hip fractures**; meta-analysis shows **increased risk of both**.
- Older adults are doubly exposed: most affected, and **ageing erodes coordination, strength, vision and cognition** — hazardous when hurrying to the toilet in the dark. **Weigh fall risk** in them and other high-risk groups.

Beyond Bother—Nocturia, Morbidity, and Mortality

- Nocturia may be a hallmark of **cardiovascular, respiratory, renal, hepatic, endocrine, neurological, psychiatric or immunological** disease.
- **Evidence is strongest for cardiometabolic and vascular disease**: hypertension, diabetes mellitus, hyperlipidaemia, obesity, metabolic syndrome, coronary artery disease, arrhythmias, cerebrovascular disease, stroke and post-stroke sequelae.
- **Mortality**: in a nationally representative US sample, **≥2 voids per night** was associated with **worse survival than 0–1**, particularly in those **<65 years**; meta-analysis shows **increased mortality in both sexes**.
- **Causality is not established**; the link is likely mediated by **falls, poor sleep and underlying comorbidity** (falls history further raises mortality risk).

KEY POINTS: RATIONALE FOR EVALUATION AND MANAGEMENT

- **Common** (highest in older adults), **bothersome**, the **commonest cause of adult sleep disruption**; hits **deep sleep**.
- Linked to **falls, fractures and systemic disease**; management matters beyond bother.

Concept — Nocturia matters because of **when it strikes and what it signals**. The first void falls in the early, restorative slow-wave sleep, and the symptom may be the visible edge of systemic disease. Its harm is therefore out of proportion to the number of voids.

CAUSES OF NOCTURIA: "SLEEP CALM" AND THE BLADDER DIARY

- Extensive differential. The relationship with poor sleep is **bidirectional**: sleep studies of patients with disordered sleep (no urinary complaint) find **bathroom trips at the time of awakening**.
- Nocturia may share a pathophysiological pathway with **systemic disease**; **lifestyle, lower urinary tract dysfunction (LUTD) and medications** may cause or aggravate it.
- **Idiopathic nocturia** is small but real. **Most real-world cases are caused by one or more of the five Sleep CALM factors**.

Bladder Diaries: Connecting Causal Factors to Clinical Symptoms

- The Sleep CALM factors look heterogeneous, but all disrupt the balance between **nocturnal urine production, functional bladder storage capacity and normal sleep architecture**. The **bladder diary** — a guideline **cornerstone** — reveals this relationship objectively.
- **Urine production is continuous**. Even after emptying at bedtime, volume climbs towards maximum capacity; when it is reached there are **only two endpoints: wake to void, or enuresis**.
- So sleeping through requires that **nocturnal production not exceed maximum functional capacity**. A production-storage mismatch arises from **excess volume (polyuria)** and/or **insufficient capacity**.

- **Sleep architecture is the third arm:** primary sleep disorders produce voids **independent of bladder volume** ("convenience" voids while already awake). Disordered sleep may also **increase nocturnal production** and/or **lower the arousal "set-point"** at which bladder volume overcomes sleep.

Diary technique

Disposable collection container → record **time and volume of every void** → record **time, volume and type of every drink** → mark **time going to bed to sleep and time rising** (separates day from night) → record **bladder sensation** (urgency) and **incontinence episodes** for each void.

Frequency-volume chart	Bladder diary
At minimum, micturition times and voided volumes only	Also requires fluid intake, pad usage and incontinence episodes

- The two terms are often used interchangeably, but consensus terminology distinguishes them as above.

Figure 112.2 — International Consultation on Incontinence (ICI) bladder diary (Bright et al., 2014). Validated **3-day** diary with columns for time, drinks (amount and type), urine output (mL), bladder sensation and pads; patients write BED and WOKE, tick an unmeasured void and write LEAK for leakage. **Bladder sensation codes:** **0** no need, voided for "social reasons"; **1** normal desire, no urgency; **2** urgency that passed before reaching the toilet; **3** urgency, reached the toilet without leaking; **4** urgency with leakage.

From the guidelines

- **EAU Female LUTS:** bladder diary as part of the **standardised assessment of female LUTS** (strong; level of evidence [LoE] 2b).
- **EAU Male LUTS:** bladder diary for male LUTS with a **prominent storage component or nocturia** (strong; LoE 3).

Nocturia Severity and Sleep

- "Going to bed intending to sleep" and "rising intending to rise" **bookend the main sleep period**; voids between them = **ANV**.
- **Intent governs:** the last void of the night does not end the sleep period if the patient stays in bed intending to sleep, even if sleep never returns. A void at or just after the end of the period is the **first morning void (daytime frequency)**.
- **Diary-based ANV is a gold standard for nocturia severity.** Symptom questionnaires correlate only modestly with diaries: respondents more often **overestimate**, especially with severe symptoms, while those with low frequency may **underplay** it.
- Interview-reported severity correlates with the diary in some populations, but **inconsistently**, and **less well as ANV rises**; overestimation at standardised interview persists **however the question is phrased**.
- **FUSP is the most important sleep measure in the diary.** It indexes how long the patient stays in **early slow-wave sleep** uninterrupted.
 - FUSP is **inversely related to the number of night-time voids**.
 - **Prolonged FUSP marks successful treatment**, with better whole-night sleep and daytime function; improved **blood glucose** has been reported after FUSP prolongation.
 - Hence FUSP is an **important outcome measure and treatment target**.
- **Total sleep duration** cuts both ways: nocturia can **truncate** sleep (patient gets up for the day after a late void), and treatment that lengthens FUSP lengthens total sleep; conversely, impaired efficiency may **keep patients in bed longer** — longer total sleep (>8 h) is more common in those with ≥2 voids. Total sleep time is **needed for calculations** but has **limited clinical utility** on its own.

Nocturnal Urine Production

- **NUV = all nocturnal voids + first morning void.** Rationale: the **untested assumption** that people void before bed and shortly after waking, so first-morning urine was made during sleep.
- **NPi = NUV/TUV** (often $\times 100$ as a percentage); **NUP = NUV/hours of main sleep period.**

COMPARISON — DEFINITIONS OF NOCTURNAL POLYURIA

Criterion	Threshold	Status	Strength / weakness
NPi	>20–33%, age-dependent (higher cut-off in older adults)	ICS; most widely used	Age bands undefined; cut-offs not empirically supported; poor discrimination
NUP	>90 mL/h	ICS-accepted alternative; favoured by many clinicians	Derived from Krimpen data; more specific for production-driven nocturia ; needs validation, especially in women
NPi (stringent)	>53%	Not ICS-approved	Proposed to avoid over-estimating NP prevalence

- **Krimpen study (landmark; community-dwelling older Dutch men):** NPi >33% in **92% of men with ≥ 2 voids but also 70% of men without nocturia** — the reason NPi discriminates poorly.
- **No universal NP cut-off exists.** Whatever the definition, **excess nocturnal production contributes to most adult nocturia**, increasingly so as severity rises.
- **Global polyuria:** >40 mL/kg/24 h (e.g. >2.8 L at 70 kg); inordinate output by day as well as night. A **relatively infrequent** cause of nocturia, but its differential is rich in **serious disease** — neoplastic, traumatic, inflammatory, immunological, psychiatric disorders of water homeostasis. **Always investigate global polyuria as a potential harbinger of disease beyond the urinary tract.**

Functional Bladder Storage Capacity

- Small functional capacity may be **night-time-specific or span 24 h** (mirroring NP versus global polyuria).
- True capacity = **voided volume + post-void residual (PVR)**; in practice **NMVV** (nocturnal) and **24-h MVV** are reasonable proxies.
- **No consensus** on diminished capacity: normal usually >300–400 mL; 150–300 mL recognised as abnormal in recent literature.

Quantifying the Relationship Between Nocturnal Urine Production and Storage

- Urge-driven waking depends on the **relationship** between production and capacity, not either alone.
- **Ni = NUV/24-h MVV** — production expressed as a **function of the patient's own capacity**. **Ni >1** = NUV exceeds capacity; higher = greater mismatch. **Ni correlates more strongly with nocturia frequency than production or capacity alone.**
- **NBCi** quantifies the gap between the patient's own maximum capacity and the **volumes at which night voids actually occur**.
 - **PNV = Ni – 1**: the voids expected if every void occurred exactly at maximum capacity. **One is subtracted because the last nocturnal volume leaves as the first morning void**, which is not a nocturia episode.
 - **NBCi = ANV – PNV.**

Worked example: NUV 800 mL, MVV 200 mL $\rightarrow$ **Ni = 4** $\rightarrow$ **PNV = 3** (three night voids of 200 mL, plus 200 mL at first morning void) $\rightarrow$ diary shows **6** night voids $\rightarrow$ **NBCi = 6 – 3 = 3.**

- **NBCi = 0**: every night void at full capacity. **NBCi >0**: voids occur below MVV; higher values = greater discordance.

- **Investigate diminished nocturnal capacity when NBCi >1.3; NBCi >2 is strongly associated with severe nocturia.**

Small capacity across 24 h	Night-time-specific small capacity
All voided volumes consistently small	Small night voids despite normal (or much larger) day voids
NBCi may be normal (MVV itself is small)	NBCi elevated

- **Convenience voids** (little or no urgency on the diary — woken by something other than urge) partly explain night-specific small capacity; they are **significantly smaller** than urge-driven voids.
- But convenience voiding is **uncommon** in the general population and remains **infrequent even in clinic patients with depression, anxiety or post-traumatic stress disorder (PTSD)**. Night-specific diminished capacity is a **small but important** entity.
- **NP, global polyuria and small capacity are not mutually exclusive.**
 - **Classical NP:** high night production **offset by low day production**, so 24-h volume is normal.
 - **Current ICS NP criteria** (NPi >33% or NUP >90 mL/h) **do not preclude concurrent global polyuria** (>40 mL/kg/24 h).
 - A small nocturnal capacity may coexist with daytime capacity that is normal, increased, or small but further reduced at night.
- **Capacity has no bearing on the polyuria definitions**, and small capacity is diagnosed **independently of urine volume**. Whether high diuresis alters perception of filling is **undemonstrated**.
- **Multiple diary abnormalities are common**, and **more so as nocturia frequency rises**.

Bladder Diaries: Practical Considerations

- **Standard of care** in all guidelines, yet **under-used**: completed by fewer than half of patients seen by urologists and about a third seen by general practitioners.
- **Adherence** is the pervasive obstacle: interference with lifestyle and routine ("too busy", "took too long", "at work that day"). In a urogynaecology sample, **unemployment and fewer children** independently predicted completion — professional, family and social commitments make diaries cumbersome.

COMPARISON — RECOMMENDED DIARY DURATION

Guideline	Duration
EAU Female LUTS	3 days
EAU Male LUTS	At least 3 days
ICS nocturia consensus	No consensus (some members favour a shorter period)

- **Longer diaries (3–7 days) are more reliable and less variable than 24-h diaries**, but burden rises with duration → adherence falls: "diary fatigue".

The nocturnal-only bladder diary

- Patients record only **night voids and the void on waking** — proposed to reduce burden and improve adherence.

Obtainable from a nocturnal-only diary	Requires 24-h data
Nocturia frequency, FUSP, duration of main sleep period, NUV, NUP, NMVV	TUV, 24-h MVV, NPi, Ni, NBCi

Obtainable from a nocturnal-only diary	Requires 24-h data
NP by NUP >90 mL/h; small nocturnal capacity; modified nocturia index	Global polyuria

- **Modified nocturia index (NUV/NMVV)** uses night-only data and is a **reasonable surrogate for the gold-standard Ni** — the strongest diary determinant of severity and treatment response.
- **Chief limitation: global polyuria cannot be diagnosed**, and it is the diagnosis most tied to serious systemic disease.
- **Partial work-around:** expressed hourly, the global polyuria threshold (40 mL/kg/24 h) is **more stringent than the NP threshold (90 mL/h) for most adults**, and the gap widens with weight — **90 mL/h at 54 kg, 117 mL/h at 70 kg, 167 mL/h at 100 kg**. So very high nocturnal volumes, particularly in heavier adults, may signal **24-h polyuria**.
- **Status:** data are **preliminary; not recommended for women by EAU Female LUTS**, although the same guideline notes acceptable sensitivity and specificity for most parameters. Reasonable **only when a patient truly cannot measure daytime volumes** — **as an alternative to no diary**.

KEY POINTS: CAUSES OF NOCTURIA

- Sleep CALM factors act on **production, storage and sleep**; the diary quantifies them, and **FUSP** tracks sleep quality and response.
- **NP in most; global polyuria** uncommon but serious; **Ni** is the strongest severity determinant.
- **3–7-day diaries** more reliable but less adhered to; night-only diaries unvalidated.

Concept — Every cause of nocturia works through **one equation: production against capacity, gated by sleep**. The diary measures both sides, and the **ratio (Ni)** predicts severity better than either side alone.

EVALUATION AND MANAGEMENT

- Diaries define the precise production–capacity–sleep relationship. **Sleep CALM** organises the initial evaluation: **history, comprehensive examination and basic testing in all** → **individually tailored adjunctive testing and multidisciplinary care** as indicated (Table 112.2).
- Guidelines support a **validated disease-specific questionnaire** at baseline and for monitoring; none is preferred. **TANGO** ("Targeting the individual's Aetiology of Nocturia to Guide Outcomes") covers the Sleep CALM factors and aids multidisciplinary assessment.
- Because Sleep CALM factors **cause** nocturia by the same mechanisms, **first-line management is patient-specific optimisation of every Sleep CALM abnormality found**.

Figure 112.3 — **TANGO short-form (TANGO-SF) screening questionnaire** (Bower et al., 2017). Tick-box statements in four domains:

- **Cardio/metabolic:** ankle/leg swelling; "fluid tablets"; kidney disease; antihypertensives; postural dizziness; high blood sugar or diabetes; unstable glucose.
- **Sleep:** **≤5 h sleep**; bad sleep quality; **>30 min to fall asleep**; waking because of the bladder; night pain; loud snoring or apnoeas.
- **Urinary tract:** voiding **within 3 h of going to sleep**; urgency most days; urgency leakage weekly or more; straining; enlarged prostate (men).
- **Wellbeing:** health "not good"; trouble staying awake driving, eating or socialising; **fall in last 3 months**; anhedonia.

From the guidelines

- **EAU Male LUTS panel on nocturia (grade A, LoE 4) and EAU Male LUTS (weak, LoE 4): treat underlying causes** — behavioural, systemic, sleep disorders, LUTD or a combination.
- **EAU LUTS guideline:** evaluation should include history and examination with attention to **sleep disorders, fluid balance, associated LUTS, cardiovascular and endocrine comorbidity, renal disease, current medications and urological history.**

TABLE 112.2 SLEEP DISORDERS, COMORBIDITIES, ACTIONS, LOWER URINARY TRACT DYSFUNCTION, AND MEDICATIONS ("SLEEP CALM") DIAGNOSTIC CONSIDERATIONS IN THE EVALUATION AND MANAGEMENT OF NOCTURIA

Factor	History	Examination	Basic testing	Ancillary testing (as indicated)	Questionnaires (as indicated)
Sleep disorders	Sleep habits, apnoea symptoms, restless legs, chronic pain, psychiatric history (anxiety, depression, PTSD, nightmares, ruminations)	—	—	Polysomnogram, home sleep apnoea test, sleep specialist, psychiatry	Sleep diary; STOP-BANG; Nocturia, Nocturnal Enuresis and Sleep-interruption Questionnaire (NNES-Q)
Comorbidities	Full medical history: cardiac, respiratory, vascular, renal, endocrine, neurological	Blood pressure; cardiopulmonary and limb examination for volume overload ; neurological	Basic metabolic panel	Waist circumference, glycated haemoglobin (HbA1c), lipids, ECG, echocardiogram, brain natriuretic peptide, albumin:creatinine ratio, renal ultrasound, calcium, thyroid function, parathyroid hormone, water deprivation or hypertonic saline test , specialist referral	TANGO
Actions	Fluid pattern, alcohol, caffeine, diet, sodium	Body mass index (BMI)	—	—	Fluid intake diary; food frequency questionnaires
LUTD	Concurrent LUTS and drugs, prior genitourinary surgery, dysuria, pelvic pain, reproductive history	Genital/pelvic; abdominal for suprapubic fullness; rectal when indicated	Urinalysis; PVR	Prostate-specific antigen (PSA), prostate sizing, cystoscopy, uroflowmetry	LUTS Score (LUTSS); OAB Symptom Score (OABSS)
Medications	Identify contributory drugs	—	—	—	—

Used with permission from Monaghan et al., Neurourol Urodyn 2023.

- **Purpose:** makes the multifactorial work-up **systematic** — each letter mapped to history → examination → test.
- **Pattern:** **basic tests are few** (metabolic panel, urinalysis, PVR); the ancillary list is **targeted, not routine**; medications need only a history.

- **Trap:** the urological column (PSA, flow, cystoscopy) is only one row of five; **starting and ending there misses four-fifths of the differential.**

Sleep CALM: Sleep Disorders

- Relationship is **bidirectional**: urge-driven waking damages early deep sleep; conversely **obstructive sleep apnoea (OSA)**, **insomnia**, **restless legs syndrome** and **parasomnias** may themselves underlie nocturnal toileting.
- **OSA** (sudden cessation of breathing from airway obstruction) is the best-described link and a **common cause of nocturia**. OSA incidence rises with nocturia frequency; severe sleep-disordered breathing → more nocturia; **high nocturia frequency predicts severe OSA** in some studies — hence calls to add nocturia to OSA screening questionnaires.

OSA — HOW IT ACTS ON ALL THREE ARMS, AND HOW CPAP REVERSES EACH

Arm	OSA mechanism	Effect of continuous positive airway pressure (CPAP)
Production (main pathway)	Obstructed breathing → hypoxia → pulmonary vasoconstriction → ↑ right atrial transmural pressure → ↑ atrial natriuretic peptide (ANP) → renal sodium and water excretion → ↑ NUV	↓ nocturnal urine production — an important mediator of response
Capacity	Intermittent hypoxia → oxidative stress → bladder ultrastructural/functional change → detrusor instability ; irritative symptoms commoner in severe OSA	Small urodynamic study: better compliance, less detrusor overactivity
Sleep	More light sleep (N1, N2), less slow-wave sleep	↑ slow-wave sleep

- **CPAP response is large:** improvement by **up to about two voids per night** in some with moderate-to-severe OSA — **among the largest effects of any nocturia intervention** in any subpopulation.
- Insomnia and other sleep disorders: mechanisms **less well described**, but likely the same three-arm effects.
- **History must establish the patient's primary motive for getting up** (Table 112.3). The **sleep diary** is under-used in urology; the authors recommend the **National Sleep Foundation Sleep Diary** (free from its website; Figure 112.4).
- **STOP-BANG:** validated OSA screen; **highly sensitive for moderate-to-severe OSA**; needs only history, vital signs and **neck circumference** (Table 112.4).

TABLE 112.3 ADDITIONAL QUESTIONS WHERE AN UNDERLYING SLEEP DISORDER IS SUGGESTED BY INITIAL SCREENING

- **Insomnia:** difficulty falling or staying asleep; daytime functioning.
- **OSA:** snoring with choking arousals; partner-witnessed apnoeas; morning headache.
- **Restless legs syndrome / periodic limb movements in sleep (RLS/PLMS):** character of sensation; **worse later in the day/evening; relieved by movement; returns minutes after sitting or lying down**; partner reports twitching or kicking.
- **Parasomnias:** odd night events (walking, screaming, eating, sexual activity); age at onset; recall; dreaming if woken; eyes open; time of night; staying in or leaving bed; injury to self or others; true nightmares (terrifying rather than merely anxious dreams).

Used with permission from Smith et al., Eur Urol Focus 2022 (PLANET study).

- **Purpose:** turn a positive sleep screen into a **working diagnosis and referral**; the RLS questions track its diagnostic features.

TABLE 112.4 THE STOP-BANG QUESTIONNAIRE FOR OBSTRUCTIVE SLEEP APNEA

Letter	Item	Positive if
S	Snoring	Loud — heard through closed doors
T	Tired	Often tired, fatigued or sleepy by day
O	Observed	Witnessed stopping breathing in sleep
P	Pressure	Hypertension, or treated for it
B	BMI	>35 kg/m ²
A	Age	>50 years
N	Neck circumference	>40 cm
G	Gender	Male

Used with permission from Chung et al., Anesthesiology 2008.

- **Scoring:** ≥3 "yes" = high risk of OSA; <3 = low risk.
- **Trap:** the numeric cut-offs — BMI 35, age 50, neck 40 cm.

Melatonin and behavioural treatment of insomnia

- **Melatonin** (pineal hormone regulating the sleep cycle) may be a **primary treatment for some subpopulations**, particularly **older adults**. Age correlates with both disordered sleep and nocturia, and older adults are **most vulnerable to sedative-hypnotics, anticholinergics and antidiuretic therapy**.
- Evidence for melatonin is **limited**, with **modest results** in some subpopulations; defining ideal candidates is a pressing research need.
- **Behavioural treatment of insomnia:** individualised sleep-hygiene education improved sleep in older adults with chronic insomnia (small randomised trial); participants with baseline nocturia also had **fewer nocturnal voids**. But **nocturia dampened the sleep-quality benefit**.
- So an **integrated cognitive-behavioural intervention for insomnia with nocturia** has been proposed by a multidisciplinary sleep panel; efficacy is under investigation.
- Know **when to involve sleep experts** in a multimodal plan.

From the guidelines

- **EAU Female LUTS (strong, LoE 1a):** women with nocturia and a history suggestive of OSA → **refer to a sleep clinic for CPAP assessment**.
- **EAU Male LUTS panel on nocturia (grade C, LoE 2):** **sleep-promoting agents** may be used to aid return to sleep.

Concept — OSA is the model for the whole chapter. **One disorder raises production (ANP), lowers capacity (detrusor instability) and fragments sleep**, which is why treating the sleep disorder yields one of the largest nocturia responses of any intervention.

Sleep CALM: Comorbidities

- Nocturia is intertwined with **cardiovascular, endocrine, renal, hepatic, neurological and immunological** disease; nocturia management is being studied **within** cardiology, endocrinology, neurology and chronic kidney disease care.
- These conditions cause nocturia chiefly through **excess urine production**. There is growing evidence that **atherosclerosis-mediated bladder vascular damage and ischaemia reduce functional capacity**. **Sleep disorders are highly prevalent** in medical disease — **comorbid insomnia may be commoner than primary insomnia**.

Figure 112.5 — PLANET (Planning Appropriate Nocturia Evaluation and Treatment) mechanisms. Five system groups — **sleep, neurological, endocrine, cardiovascular, renal** — act through relative lightening of sleep, **dysregulated antidiuresis, natriuresis**, excessive water or salt intake, **salt and water retention, nocturnal hypertension, nocturnal overfiltration, impaired urine concentration** and **reduced bladder control**.

- **Nocturia as an early sign:** of **unrecognised hypertension**; a **precursor of metabolic syndrome**; an independent predictor of **coronary heart disease in younger men** and **cardiovascular death in older men**; severity is associated with subsequent **atherosclerotic cardiovascular disease** (myocardial infarction, stroke, coronary death).
- May flag **end-organ damage** in diabetes, **albuminuria/early kidney disease**, end-stage renal disease and **hepatic synthetic dysfunction in cirrhosis**. As a possible **sole presenting sign of severe illness**, nocturia matters for public health.
- **Work-up:** comprehensive history with review of systems, **examination beyond the genitourinary tract**, and a **low threshold for primary or specialist referral**.
- **EAU Male LUTS:** consider **systemic causes in all cases**; multidisciplinary care is best practice when indicated. **Even non-bothersome nocturia merits evaluation**, including in patients presenting for other reasons.

Figure 112.6 — PLANET assessment: SCREeN (Sleep medicine, Cardiovascular, Renal, Endocrine, Neurology).

Domain	Conditions	Screening questions
Sleep	OSA, insomnia, periodic limb movements, parasomnias	Problems sleeping aside from voiding? Gasp or stop breathing? Wake refreshed? Fall asleep by day?
Cardiovascular	Congestive cardiac failure, hypertension	Breathless on exertion? Ankle swelling?
Renal	Chronic kidney disease	(Investigations)
Endocrine	Diabetes mellitus, thyroid dysfunction, diabetes insipidus, pregnancy, menopause	Excessive thirst? Hand tremor? (Women) change in periods?
Neurological	All neurological conditions	Light-headed on standing? Slowness of movement? Problems controlling legs?

- **Initial investigations:** **bladder diary**; blood for **renal and thyroid function, calcium, HbA1c**; urine dipstick for **blood, protein** and **albumin:creatinine ratio (ACR)**.
- **Treating the disease improves nocturia:** key examples are heart failure, hypertension and disorders of water homeostasis.

Volume Overload

- By day, intravascular fluid "**third-spaces**" into the interstitium and collects in **dependent legs**; it is **not excreted by day** but **mobilised when lying down**, adding to NUV — even in healthy adults.
- **Exaggerated in hypervolaemic states** such as congestive heart failure, of which third-spacing is cardinal.
- Acute decompensated heart failure: more nocturia before admission ↔ **more leg oedema and greater NUV**; **diuretic treatment** reduced nocturia alongside oedema and NUV. Similar findings in cardiology outpatients with overt overload treated to best practice.

Hypertension

- Normal **circadian dip of 10–20%** in blood pressure during sleep. **Non-dipping hypertension** occurs in **up to 40–50% of hypertensives** and independently predicts target-organ damage and adverse cardiovascular outcomes.
- Non-dipping probably mediates the nocturia–hypertension link: nocturia is independently associated with **higher night-time pressure and blunted dipping**.

- **Mechanism:** renin–angiotensin–aldosterone dysregulation → **excess aldosterone** → **nocturnal natriuresis**; aberrant sympathetic activity and natriuretic peptides contribute.
- **Optimising hypertension may help nocturia.** Hydrochlorothiazide (HCTZ) is a potent antihypertensive with a **more pronounced antihypertensive but more modest diuretic effect than loop diuretics** such as furosemide.
 - Pilot study: HCTZ 25 mg improved **nocturia, NUV and NPi** in men with terazosin-refractory LUTS.
 - Adjunctive HCTZ 12.5 mg in elderly losartan-refractory hypertension **did not worsen nocturia**.
 - **Bedtime antihypertensive chronotherapy** trials support better circadian control and fewer major cardiovascular events, but **did not measure nocturia**.

Disorders of Water Homeostasis

Arginine vasopressin (AVP) physiology: made in the **paraventricular and supraoptic nuclei** → transported to the **posterior pituitary (neurohypophysis)** → released to osmotic and volume stimuli (**rising serum osmolality is the main stimulus**) → binds **V2 receptors on collecting-duct principal cells** → ↑ water permeability → water reabsorbed down osmotic gradients until isotonic.

High AVP	Low AVP
Water conserved; small volume of concentrated urine	Relatively large volume of dilute urine

- Impaired AVP secretion or action shows on the diary as **24-h excess production** — so **global polyuria must always raise suspicion of a disorder of water homeostasis**.

COMPARISON — THE POLYURIA–POLYDIPSIA SYNDROME (LARGE VOLUMES OF DILUTE URINE)

Condition	Defect	Causes
Central diabetes insipidus (DI)	Insufficient AVP secretion	Central nervous system (CNS) tumours, haemorrhage, trauma, autoimmune disease
Nephrogenic DI	Renal insensitivity to AVP	Some antifungals, antibiotics, steroids, chemotherapeutics; transient in renal failure or major electrolyte disturbance
Primary polydipsia	Physiological suppression of AVP by excess intake	Idiopathic; schizophrenia, anxiety, other psychiatric illness

- Polyuria is an infrequent cause of nocturia, but **every patient with global polyuria must be evaluated** so as not to miss a traumatic, neoplastic, inflammatory, immunological or psychiatric process.
- **Diagnosis is changing: arginine-stimulated copeptin** (a cleavage product of prepro-vasopressin) appears **superior to the traditional overnight water deprivation test**.
- Cause-specific treatments restore water homeostasis, **normalise output and probably improve nocturia**.

From the guidelines

- **EAU Male LUTS panel on nocturia:** nocturia may have a **significant medical cause** — an opportunity to screen for **undiagnosed or poorly managed disease**, reduce complications and save cost.
- **EAU Male LUTS:** if systemic factors or sleep disorders may be important, **involve appropriate medical expertise** — where a known condition is sub-optimally managed or signs suggest an undiagnosed one.
- **EAU Male LUTS:** therapy may be driven by bother, but **non-bothersome nocturia may warrant a frequency–volume chart** because a serious underlying condition must be considered.

Concept — Comorbid disease reaches the bladder mostly **through the kidney**: third-spaced fluid, nocturnal natriuresis and failed antidiuresis all raise night production. That is why **global polyuria and non-bothersome nocturia are both reasons to look beyond urology**.

Sleep CALM: Actions

- Assess **excess fluid, caffeine and alcohol**, particularly in the **3–4 h before bed** (Figure 112.7).
- **Epidemiology is inconsistent** — large cohorts show no clear association, and some find nocturics *less* likely to take evening caffeine. This probably reflects **cross-sectional design: patients self-regulate intake** as self-treatment.
- **Interventional evidence is thin:**
 - Crossover trial in OAB: **25% fluid reduction** reduced nocturia, urgency and daytime frequency, but the nocturia gain was of **modest clinical significance**.
 - Men with **24-h output >30 mL/kg** guided to **<30 mL/kg**: about one fewer night void, correlating with the fall in nocturnal volume; no adverse events.
 - Bottom line: benefit varies; **patients with higher baseline intake gain most**.
- **Physiology is the stronger evidence:** urine output tracks total intake.

TIMING OF DIURETIC STIMULI RELEVANT TO BEDTIME

Stimulus	Effect	Timing
Acute water bolus (a glass over 15 min)	Mostly excreted; plasma AVP suppressed → brisk water diuresis	Onset ≈1 h after ingestion
Caffeine; alcohol	Acute diuresis; both also disturb sleep architecture	Within the first 2–3 h
Caffeine clearance	Adult half-life, with wide population variation	2.5–4.5 h — earlier cut-off for heavy users
Protein-rich evening meal (amino acid bolus)	Sustained glomerular hyperfiltration	2–3 h

- **Practice:** fluid advice is physiologically sound and **rational in many patients** — centre it on the evening, with **aggressive restriction from 3–4 h before intended sleep**.
- **Cautions:** disorders of water and electrolyte homeostasis; **older adults** — compensate with more fluid by day to avoid **dehydration**, common and harmful in this group.
- **Dietary sodium:** high intake is an **independent risk factor**; **restriction is effective** in several settings (low-quality evidence); mechanism under investigation.
- **Obesity** is associated with nocturia in some populations; **regular exercise** with lower risk. **Bariatric surgery** improves nocturia (not a primary treatment) — underlining the relevance of BMI.
- **Compression stockings:** counter daytime leg fluid accumulation, reducing night mobilisation — a **potentially effective** intervention.
- Other conservative options as indicated: **mindfulness, bladder retraining, pelvic floor muscle training, treating constipation**.

Figure 112.7 — *Conservative interventions for nocturia.* Sleep hygiene; evening fluid restriction; less caffeine and alcohol; dietary sodium limitation; bladder and pelvic floor training; exercise and weight loss; compression stockings.

- **EAU Male LUTS (weak, LoE 4):** discuss **behavioural changes** to reduce nocturnal volume and episodes and improve sleep.

Concept — *Lifestyle advice rests on **diuretic kinetics rather than trial data**. Anything drunk or eaten within the few hours before bed is excreted during sleep, so timing matters more than total amount.*

Sleep CALM: Lower Urinary Tract Dysfunction

- Nocturia is often isolated, but is also a component of **bladder outlet obstruction (BOO)**, **OAB**, **detrusor underactivity**, **neurogenic bladder**, **dysfunctional voiding** and the **genitourinary syndrome of menopause (GSM)**. It is commoner in men with clinically significant BPH and is present in **up to 85% of OAB trial participants**.
- Historically nocturia was **just one LUTS attributed to BPH** (and a core OAB feature) long before it was studied in its own right.
- **Isolated nocturia is not a LUTS and would not be expected to respond to lower urinary tract drugs.** Those drugs — historically the backbone of treatment — **seldom give clinically meaningful benefit**, but remain reasonable **when LUTD is justifiably suspected**.

INITIAL LUTD ASSESSMENT IN NOCTURIA (EAU, ICS)

Component	Recommendation
History	Concurrent LUTS; disease-specific questionnaires may help
Examination	Abdominal (bladder fullness) + genital and/or digital rectal examination (DRE)
Urinalysis	All patients ; if urinary tract infection (UTI) found → treat → repeat urinalysis
PVR bladder scan	Both women and men
PSA	Only if the result would change decisions or treatment

- LUTD (incomplete emptying, abnormal afferent signalling) typically gives **smaller voided volumes** — nocturia from **diminished functional capacity**.
- **Neurological patients**: nocturia highly prevalent but under-studied; manage by **neurogenic LUTD guidelines**.

From the guidelines

- **EAU Female LUTS**: urological causes with prominent nocturia include **OAB**, **BOO**, **detrusor underactivity** and **dysfunctional voiding**.
- **EAU Male LUTS**: nocturia can be part of OAB and chronic pelvic pain syndrome; **with BOO it is debated whether the link is causal or coexistence**. Behavioural, sleep and systemic causes may coexist and must be considered in all cases. **Only where LUTD contributes should nocturia be termed a LUTS**.

Benign Prostatic Obstruction

- **Relieving BOO should help nocturia** by lowering PVR, increasing functional capacity, reducing global frequency and **reducing afferent input from the bladder neck and prostatic urethra**.
- **α -blockers**: inhibit noradrenaline at prostatic smooth muscle → ↓ prostatic tone and outlet resistance. **α 1-receptors also lie in bladder, urethra and CNS**, and α -blockers reduce CNS afferent activity. Available: **alfuzosin, doxazosin, tamsulosin, terazosin, silodosin**.
- **5 α -reductase inhibitors (5ARIs)**, **dutasteride and finasteride**, block formation of serum and intraprostatic **dihydrotestosterone (DHT)** → prostatic involution and ↓ volume.
- Nocturia data come almost entirely from **secondary or post hoc endpoints in BPH/LUTS trials**, in which nocturia did not determine eligibility.

COMPARISON — PROSTATIC PHARMACOTHERAPY TRIALS REPORTING NOCTURIA

Trial / analysis	Arms	Nocturia endpoint	Verdict
Pooled placebo-controlled silodosin trials (post hoc)	Silodosin 8 mg vs placebo	≥1-void improvement if ≥2 voids	Statistically superior, but about half of placebo patients also improved
Silodosin vs tamsulosin RCT	Silodosin vs tamsulosin vs placebo	Change in frequency	Silodosin, not tamsulosin , beat placebo; absolute differences a fraction of a void
Tamsulosin controlled-absorption pilot (men with ≥2 voids)	Tamsulosin vs placebo	FUSP (primary); frequency	FUSP and frequency improved in both arms; no significant difference
Pooled phase III dutasteride trials	Dutasteride 0.5 mg vs placebo	Change at 2 years	Significant but small
REDUCE (landmark; chemoprevention)	Dutasteride 0.5 mg vs placebo	Change at 2 and 4 years	Nocturia improved by -0.15 and -0.24 voids ; no sleep-quality benefit
VA Cooperative Study (landmark)	Terazosin 10 mg, finasteride 5 mg, combination, placebo	Frequency at 1 year (baseline 2.5)	1.8 terazosin ; 2.1 finasteride; 2.0 combination; 2.1 placebo — only the α-blocker separated
CombAT (landmark, 4 years)	Tamsulosin 0.4 mg, dutasteride 0.5 mg, combination	Change in frequency	Combination -0.5 vs tamsulosin -0.3 and dutasteride -0.4 (both $P \leq 0.01$)
MTOPS (landmark)	Doxazosin 8 mg, finasteride 5 mg, combination, placebo	Change at 1 year	Combination -0.58 and doxazosin -0.54 beat placebo -0.35; finasteride -0.40 did not; persisted at 4 years

- **Lesson: statistical without clinical significance** — fractions of a void, large placebo response.
- **Pattern:** the **α-blocker** carries most of the effect; 5ARIs alone are weakest; **combination ≈ α-blocker alone** for nocturia.
- **Trap:** quoting a significant P value as evidence of worthwhile nocturia benefit.
- **Phosphodiesterase type 5 inhibitors (PDE5Is):** PDE5 is highly expressed in the lower urinary tract and its vessels, driving prostatic, bladder and arterial tone. Inhibition ↑ blood flow and oxygenation and inhibits non-voiding contractions and mechanosensitive afferents. PDE5Is **improve overall LUTS**, but **no clinical benefit for nocturia** (pooled tadalafil trials: -0.5 vs -0.4 placebo). May be used for moderate-to-severe LUTS with or without erectile dysfunction — **but not offered for nocturia**.

Bladder outlet surgery

- Offered to appropriately selected men with BPH-related LUTS **refractory to medical therapy**.

Comparison	Finding
Real-world cohort: transurethral resection of the prostate (TURP), transurethral microwave therapy (TUMT), α-blocker, watchful waiting	Reduction in nocturia episodes ranked TURP > TUMT > α-blocker > watchful waiting

Comparison	Finding
Consecutive new BPH-LUTS patients with ≥ 2 voids: TURP vs tamsulosin	TURP reduced nocturia in a larger proportion
Small RCT: TURP vs tamsulosin 0.4 mg (untreated nocturia)	Change favoured TURP (both reached 1.4 voids at 1 year) — possibly driven by higher baseline in the TURP arm ; better nocturia QoL with TURP, no difference in uninterrupted sleep
Prospective TURP series (≥ 2 voids)	Nocturia fell at 6 months; no change in subjective sleep quality
TURP or simple prostatectomy series	Nocturia and FUSP both improved ; each independently associated with better QoL

- **Verdict:** surgery may be **slightly more effective than medical therapy** for frequency, but the **real clinical benefit is unclear** because sleep and QoL data are inconsistent.
- **Newer BPH treatments** — GreenLight photoselective vaporisation, Aquablation, Rezum, UroLift, prostatic artery embolisation — each have advantages over TURP, but **none is proven more effective than TURP for LUTS, specifically nocturia**.
- **Counsel before surgery** on the limited evidence and possible **sub-optimal nocturia response**, especially when nocturia is the most bothersome complaint.

Overactive Bladder Syndrome

- **Antimuscarinics** (historical cornerstone): **M3 receptors** mediate detrusor contraction; antagonism of post-junctional excitatory receptors → relaxation. Muscarinic receptors in the **urothelium and afferent pathways** may also contribute to OAB.
- **Proposed nocturia mechanisms:** ↓ afferent activity (less nocturnal urgency), ↑ nocturnal capacity, longer inter-void interval. Agents: **darifenacin, fesoterodine, oxybutynin, propiverine, solifenacin, tolterodine, trospium**.
- Evidence mostly secondary/post hoc; significance **inconsistent**. **Nocturnal urgency and nocturnal voids are among the most refractory OAB symptoms** — good daytime response does not predict night-time benefit.

COMPARISON — ANTIMUSCARINIC TRIALS WITH A NOCTURNAL PRIMARY ENDPOINT

Trial	Population / design	Primary endpoint	Verdict
Tolterodine extended release (ER) 4 mg vs placebo	OAB with ≥ 2.5 night voids ; placebo run-in; dosed ≥ 4 h before bed; 12 weeks; 7-day diaries with urgency rating	Change in nocturnal micturitions	Negative ; but urge-related and severe urge-related night voids fell significantly
Fesoterodine 4→8 mg vs placebo — the only trial with nocturnal urgency as primary endpoint	OAB with 2–8 nocturnal urgency episodes (Urinary Sensation Scale ≥ 3); 12 weeks	Change in nocturnal urgency episodes	Positive ; total night voids also favoured fesoterodine

- Post hoc: patients voiding **below their own 24-h capacity (elevated NBCi)** fare **better with evening antimuscarinic** than those already voiding near capacity.
- **Bottom line:** antimuscarinics help most with **severe urgency and small nocturnal voided volumes**, but as for BPH drugs, **significance seldom translates into meaningful benefit**.

- **β₃-adrenoceptor agonists (mirabegron, vibegron):** raise cyclic adenosine monophosphate in the storage phase → detrusor relaxation; OAB relief **comparable to antimuscarinics with fewer side effects**. Mirabegron gives a similar or slightly larger rise in voided volume, perhaps not clinically noticeable.
 - **Systematic review:** mirabegron 50 and 100 mg reduce nocturia **marginally versus placebo** (weighted mean differences of about -0.1 to -0.2 voids); **50 mg ≈ 100 mg**; 50 mg **slightly better than tolterodine**, 100 mg not; heterogeneous trials.
 - **Vibegron** 50 and 100 mg: modestly greater reduction than placebo (secondary analysis).
 - **No β₃-agonist study has nocturia as its primary outcome**; comparison with antimuscarinics for nocturia is **poorly characterised**.

Behavioural therapy for OAB and nocturia

- **Before pharmacotherapy, all OAB patients should be offered behavioural therapy:** bladder training, pelvic floor muscle (PFM) therapy, fluid management, bladder-control strategies.

COMPARISON — BEHAVIOURAL VERSUS DRUG THERAPY (NOCTURIA AS SECONDARY OUTCOME)

Population	Arms	Nocturia verdict
Women with urgency urinary incontinence (UUI)	Tolterodine ER ± supervised behavioural training	Trend to combination; not significant
Older women with UUI	Biofeedback-assisted PFM training vs titrated oxybutynin vs placebo	Behavioural > drug > placebo
OAB (open-label)	Darifenacin ± behavioural modification	Both improved; no difference
Untreated women with OAB	Tolterodine vs bladder training vs combination	All improved; comparison limited by baseline imbalance and attrition
Women with OAB	Trospium vs physiotherapy (interferential current, PFM, bladder training) vs combination	Physiotherapy and combination > drug alone
Male veterans, OAB despite α-blocker, no BOO	Titrated oxybutynin vs adjunctive behavioural therapy	Daytime equivalent; behavioural better for nocturia
Men with OAB (6 weeks, then all step up to combination)	Behavioural vs tolterodine 4 mg + tamsulosin 0.4 mg vs combination	Initially combination > behavioural > drug ; after step-up no difference → behavioural alone is a reasonable first-line option

- **Pattern:** behavioural therapy **matches or beats drugs** for nocturia; adding drugs adds little.
- **Caveats:** nocturia was **never the primary outcome**, and what counts as "behavioural" is **heterogeneous**.
- **Resource burden can be reduced:** unsupervised training via a **2-h class or a 20-min video** both reduced nocturia over 2 years in women; **individual and group PFM training** both improved nocturia in older women.

Third-line OAB therapy

Therapy	Nocturia evidence (secondary endpoints, 12 weeks)
OnabotulinumtoxinA 100 U (two phase III trials, OAB with UUI)	Better than placebo by a fraction of a void; no active comparator ; trials comparing it with anticholinergics did not assess nocturia
OnabotulinumtoxinA vs sacral neuromodulation (SNM) (refractory UUI)	No difference ; nocturia not assessed in other key SNM trials

Therapy	Nocturia evidence (secondary endpoints, 12 weeks)
Percutaneous tibial nerve stimulation (PTNS) vs sham	PTNS better
PTNS vs tolterodine 4 mg	Similar

- Third-line treatment probably **beats placebo** for nocturia, but **nocturia has never been a prospective primary outcome**, and superiority to drugs is unknown.

Nocturia and the Genitourinary Syndrome of Menopause

- GSM: LUTS plus genital and sexual symptoms from **reduced oestrogenic stimulation**. Nocturia is common within it. Mechanism is probably multifactorial: **LUTD + excess nocturnal production + sleep impairment**.
- **Hormone replacement therapy may help** postmenopausal women with bothersome urinary symptoms: **oral oestrogen, vaginal ring or pessary**, and several formulations in a recent pilot. The selective oestrogen receptor modulator **ospemifene** is potentially effective.

COMPARISON — GUIDELINE RECOMMENDATIONS FOR LUTD-DIRECTED THERAPY IN NOCTURIA

Therapy	EAU Male LUTS 2022	EAU Male panel on nocturia 2017	EAU Female LUTS 2022
α1-blocker	Offer for nocturia with LUTS (weak, LoE 2)	May be offered (grade B, LoE 1b)	—
5ARI	Offer with LUTS and prostate >40 mL (weak, LoE 2)	May be offered with moderate-to-severe LUTS and prostate >40 mL (grade C, LoE 1b)	—
PDE5I	Do not offer (weak, LoE 2)	Should not be offered (grade B, LoE 1b)	—
Antimuscarinic	Offer for nocturia with OAB (weak, LoE 2)	May be offered with OAB (grade B, LoE 1b)	Offer with UII or other LUTS after counselling (strong , LoE 1b)
PFM training	—	—	Offer , individual or group, with incontinence or storage LUTS (strong , LoE 1b)
Behavioural + anticholinergic	—	—	Inform: combination unlikely to beat either alone (weak, LoE 1b)
Vaginal oestrogen	—	—	Offer after counselling (weak, LoE 1b)

- **Pattern**: every male LUTD drug recommendation is **weak**; the only **strong** recommendations are in the female guideline (antimuscarinic, PFM training). **PDE5Is are the one explicit "do not offer"**.
- **Trap**: the 5ARI prostate threshold is **>40 mL**.

TABLE 112.5 PHARMACOLOGIC AGENTS FOR NOCTURIA WITH ASSOCIATED LUTS — A. DOSING AND ORGAN ADJUSTMENT

Drug	Oral dose	Renal	Hepatic / cardiovascular
α-blockers			

Drug	Oral dose	Renal	Hepatic / cardiovascular
Alfuzosin	10 mg daily	Caution if creatinine clearance (CrCl) <30 mL/min	Contraindicated in moderate–severe hepatic impairment; caution with tachyarrhythmia or ischaemia
Doxazosin	Immediate release (IR) 1–8 mg; extended release (ER) 4–8 mg	No adjustment	Not recommended in severe impairment; caution in heart failure, angina, recent myocardial infarction
Silodosin	8 mg with food	4 mg if CrCl 30–50; contraindicated <30	Caution mild–moderate; contraindicated severe
Tamsulosin	0.4–0.8 mg, 30 min after the same meal	Caution CrCl <10	No adjustment; caution in heart failure
Terazosin	1–5 mg at bedtime (max 20 mg)	No adjustment	No adjustment; caution in heart failure
5α-reductase inhibitors			
Dutasteride	0.5 mg	No adjustment	Caution (extensive hepatic metabolism)
Finasteride	5 mg	No adjustment	Caution (extensive hepatic metabolism)
Antimuscarinics			
Darifenacin	7.5–15 mg	No adjustment	Max 7.5 mg moderate; not recommended severe
Fesoterodine	4–8 mg (max 4 mg with potent CYP3A4 inhibitor)	Max 4 mg if CrCl <30	Not recommended severe
Oxybutynin	IR 5 mg two to three times daily (max four); older adults start 2.5 mg; ER 5–10 mg (max 30 mg)	Caution (limited data)	Caution (not studied)
Solifenacin	5–10 mg	Max 5 mg if CrCl <30	Max 5 mg moderate; not recommended severe
Tolterodine	IR 1–2 mg twice daily; ER 2–4 mg (lower with CYP3A4 inhibitors)	CrCl 10–30: IR 1 mg/day, ER 2 mg/day; ER not recommended <10	IR 1 mg twice daily severe; ER 2 mg mild–moderate, not recommended severe
Tropium chloride	IR 20 mg twice daily (consider 20 mg/day in elderly); ER 60 mg in the morning	CrCl <30: IR 20 mg at bedtime; ER not recommended	Caution moderate–severe
β3-agonists			
Mirabegron	25–50 mg	Max 25 mg if estimated glomerular filtration rate (eGFR) 15–30; not recommended <15	Max 25 mg moderate; not recommended severe
Vibegron	75 mg	Not recommended eGFR <15	Not recommended severe

TABLE 112.5 (CONTINUED) — B. CONTRAINDICATIONS AND SIGNATURE ADVERSE EFFECTS

Drug	Notable contraindications	Signature adverse effects / precautions
Alfuzosin	Potent CYP3A4 inhibitors (ketoconazole, itraconazole, ritonavir); other α 1-blockers	Dizziness; intraoperative floppy iris syndrome (IFIS) , orthostatic hypotension, retrograde ejaculation, rare priapism; QTc prolongation
Doxazosin	Galactose intolerance; hypersensitivity to quinazolines (prazosin, terazosin)	Dizziness, fatigue, malaise, headache ; IFIS, orthostasis, priapism; rare reversible leucopenia
Silodosin	Potent CYP3A4 inhibitors; other α 1-blockers; CrCl <30	Retrograde ejaculation — the dominant effect ; orthostatic hypotension (caution >65 years), syncope, IFIS
Tamsulosin	Strong CYP3A4 inhibitors (ketoconazole)	Headache, orthostatic hypotension , rhinitis, ejaculatory dysfunction , dizziness; rare anaphylaxis with sulfonamide allergy ; IFIS
Terazosin	Hypersensitivity	Dizziness, myasthenia ; IFIS, orthostasis, priapism
Dutasteride	Other 5ARIs; pregnancy	Impotence and ↓ libido (highest in first 6 months) , gynaecomastia; PSA halved within 3–6 months ; US Food and Drug Administration (FDA) pregnancy category X ; sexual dysfunction may persist
Finasteride	Other 5ARIs; pregnancy	Impotence , ↓ libido, ejaculatory disorder; gynaecomastia; male breast cancer (<1%) ; PSA halved within 3–6 months; category X
Antimuscarinics (class)	Gastric retention, urinary retention, uncontrolled narrow-angle glaucoma , hypersensitivity	Dry mouth, constipation ; anticholinergic effects; rare angioedema; Beers Criteria — potentially inappropriate ≥65 years
Oxybutynin	Class; severe dysmotility	Highest dry-mouth burden in the table; dizziness, drowsiness, confusion
Solifenacin	Class; dialysis	Dose-dependent QTc prolongation
Fesoterodine / tolterodine	Class; cross-hypersensitivity with each other	Tolterodine: QTc at supratherapeutic dose; CYP3A4 dose adjustment
Trospium	Class	Eliminated by active tubular secretion — caution with vancomycin, morphine, metformin, tenofovir; potentiated by alcohol/sedatives; caution in ulcerative colitis
Mirabegron	Severe uncontrolled hypertension	Hypertension (commonest), tachycardia, palpitations; caution urinary retention
Vibegron	Hypersensitivity	Headache, nasopharyngitis; ↑ PVR / retention (caution)

Information derived from Lexicomp Lexi-Drugs; limited summary — consult full prescriber information.

- **Purpose:** safe prescribing in an **old, multimorbid, often renally impaired** population.
- **Pattern:** α -blockers share **IFIS and orthostasis**; 5ARIs share **sexual effects, PSA halving and teratogenicity**; antimuscarinics share **retention, glaucoma and Beers-list** warnings; β 3-agonists swap anticholinergic burden for **blood-pressure effects**.

- **Traps:** **silodosin is contraindicated at CrCl <30** (halve to 4 mg at 30–50); **trospium is the tubular-secretion drug**; **mirabegron's signature effect is hypertension**; antimuscarinics are **Beers-list drugs in the over-65s** — the core nocturia population.

Concept — Lower urinary tract drugs act on **capacity and outflow**, but most nocturia is driven by **production**. So they give statistically significant but clinically small effects, and they help only when LUTD genuinely contributes: **treat the mechanism, not the organ**.

Sleep CALM: Medications

- Many drugs used for medical disease **cause nocturia** (Table 112.6), by ↑ nocturnal production, **insomnia/CNS effects** or **direct lower urinary tract effects**.
- **Careful medication reconciliation** is essential; establish the **time of administration**. Initial management: **adjust timing or frequency, or substitute** a drug with fewer urinary effects.

TABLE 112.6 DRUG EFFECTS CAUSING NOCTURIA

Mechanism	Drugs	How
Increased urine output	Diuretics	Diuresis
	Selective serotonin reuptake inhibitors (SSRIs)	Block antidiuretic hormone (ADH) secretion
	Calcium channel blockers	Oedema from ↓ arteriolar resistance; ↑ ANP; block proximal tubular sodium reabsorption
	Tetracycline	Attenuates ADH via ↓ cyclic adenosine monophosphate (cAMP)
	Lithium	↓ Aquaporin-2; lithium-induced oedema
	Sodium-glucose cotransporter 2 (SGLT-2) inhibitors	Glycosuria
Insomnia / CNS effects	CNS stimulants	Dextroamphetamine, methylphenidate
	Antihypertensives	α-blockers, β-blockers, methyldopa
	Respiratory; decongestants	Salbutamol (albuterol), theophylline; phenylephrine, pseudoephedrine
	Hormones	Corticosteroids, thyroid hormone
	Psychotropics	Monoamine oxidase inhibitors (MAOIs), SSRIs, atypical antidepressants
	Dopaminergic; antiepileptic	Carbidopa; phenytoin
Direct lower urinary tract effects	Ketamine	Direct bladder toxin
	Tiaprofenic acid (Surgam)	Toxic cystitis
	Cyclophosphamide	Bladder toxicity

Modified from Weiss et al., Nocturia Think Tank: focus on nocturnal polyuria, ICI-RS 2011.

- **Pattern:** the three groups mirror the **production / sleep / bladder** arms; **SSRIs appear twice** (ADH effect and sleep).
- **Traps:** **calcium channel blockers** cause nocturia through **oedema** (third-spacing, mobilised at night); **SGLT-2 inhibitors** through glycosuria; **α-blockers**, used to treat nocturia with BPH, also appear as a cause of insomnia.

Multicomponent Therapy

- Most nocturia has **more than one cause**, and the Sleep CALM factors **interact**:
 - Poor health and comorbidity burden are the **strongest predictors of sleep disturbance**; many chronic-disease drugs cause insomnia.
 - LUTS severity is linked to cardiovascular risk, inflammation and systemic disease.
 - **Older age** brings more disease, adverse sleep changes and **polypharmacy**.
- A multifactorial cause needs a **multimodal plan** — the basis of the **EAU Male LUTS "Shared Care Pathway"**, which addresses sleep, medical and lower urinary tract factors **concurrently**.
- Small trials show benefit from **multicomponent, aetiology-directed treatment**; an individualised plan addressing several Sleep CALM factors in older male US veterans improved **frequency, bother and sleep outcomes**.

KEY POINTS: EVALUATION AND MANAGEMENT

- **Optimise every Sleep CALM factor**; sleep is bidirectional — CPAP, melatonin, insomnia therapy may help.
- **Head-to-toe work-up**: nocturia may be the sole sign of disease, and treating disease helps nocturia.
- **Evening fluid, alcohol, caffeine, sodium** restriction: variable evidence, low risk.
- **LUTD drugs: modest benefit**; $\beta 3$ vs antimuscarinic unknown; **oestrogen in GSM**.
- **Reconcile medications**; **multicomponent** treatment is likely additive.

Concept — Sleep CALM is a **diagnostic checklist and a treatment plan at once**. Each factor causes nocturia through production, capacity or sleep, so identifying it is already the start of treating it. Because the factors coexist, they are **treated together, not in turn**.

ADDITIONAL TREATMENT CONSIDERATIONS

Refractory Nocturia: A Stepwise Approach

Figure 112.8 — Sleep CALM diagnostic algorithm (Monaghan et al., 2023).

History + physical examination + individualised ancillary testing → **identify modifiable Sleep CALM factors** → **optimise Sleep CALM** → **conservative management** → **pharmacotherapy in carefully selected patients** — with the **voiding diary** running alongside every step.

- **Even refractory nocturia starts conservatively**: lifestyle measures have variable evidence but **minimal risk**. Drugs are for nocturia **refractory to Sleep CALM optimisation and conservative management**.
- **In the USA, bedtime antidiuretic replacement is the only drug specifically indicated for nocturia** — for **≥ 2 voids per night due to NP**. Guidelines also discuss **timed diuretics** (e.g. early afternoon), which shift urine output into the day and reduce NUV.
- **Why target nocturnal volume even when capacity is small**:
 - Production and capacity both determine the mismatch, but **excess nocturnal production is the stronger correlate of frequency and of treatment response**.
 - NP patients benefit most; but **even patients without NP respond** to volume-reducing treatment.
 - In patients with **impaired nocturnal capacity**, production-lowering therapy helps them **void closer to capacity**. In diary-confirmed small capacity, improvement came with **falling NUV and unchanged capacity**.

- Production-lowering therapy improves nocturia in **BPH and OAB**, as monotherapy or add-on.
- **EAU Male LUTS panel on nocturia (grade A, LoE 3)**: discuss **lifestyle changes** to reduce nocturnal volume and episodes and improve sleep.

Antidiuretic Replacement Therapy

- **Endogenous AVP normally rises during sleep**, conserving water, lowering NUV and counteracting bladder filling. A **blunted nocturnal AVP rise** occurs in some NP samples — leading some to regard NP as **partial central DI**.
- AVP is **technically difficult to measure**, and its role may differ between populations; but synthetic antidiuretics predictably ↑ water permeability and reabsorption, dilute extracellular fluid and **concentrate urine**.
- **Bedtime antidiuretic** → ↓ NUV: a rational option for **refractory nocturia without other identifiable contributory factors**.
- **Desmopressin (1-deamino-8-D-arginine vasopressin, DDAVP)**: **selective V2 agonist** — antidiuretic action of AVP **without pressor activity**. Used since the 1980s; approved for nocturia in **>80 countries**.
- US-approved **low-dose formulations** for NP with **≥2 voids per night**: **desmopressin acetate nasal spray (DANS)** and **orally disintegrating tablets (ODT)**. They offer **greater bioavailability, faster onset** and a better efficacy–side-effect balance, notably for **hyponatraemia — the most clinically significant side effect**.
- Antidiuretic therapy is supported by **level 1b evidence** from multiple phase III placebo-controlled randomised controlled trials (RCTs) (Table 112.7).

COMPARISON — US LOW-DOSE DESMOPRESSIN FORMULATIONS

	DANS	ODT
FDA approval	March 2017 — first	June 2018
Special feature	Cyclic fatty-acid excipient → better absorption, more predictable activity	Sex-specific dosing
Doses	0.83 and 1.66 mcg/0.1 mL (= 0.75 and 1.5 mcg free base)	27.7 and 55.3 mcg (= 25 and 50 mcg desmopressin)
Dose selection	1.66 mcg if <65 years with no hyponatraemia risk; ≥65 years or at risk → start 0.83 mcg , increase if sodium normal	Women 27.7 mcg; men 55.3 mcg
Administration	One spray in either nostril ≈30 min before bed	Sublingual, without water , ≈60 min before bed

- **Pattern**: DANS stratifies dose by **age/risk**; ODT stratifies by **sex**.
- **Trap**: the labelled ODT strengths (27.7/55.3 mcg) are **equivalent to 25/50 mcg desmopressin**; trial doses are quoted as 25/50 mcg.

Sodium Monitoring

- **Hyponatraemia risk** is most strongly associated with **older age**, particularly with **low baseline sodium**, and is **dose-dependent**. It was seen mainly with **earlier high-dose oral tablets** in older patients (sodium <130 mmol/L in up to about 5%). Risk with DANS or ODT is **lower**.
- **Low-dose rationale**: the **median effective dose** (therapeutic response in 50%) is **far lower than the dose for maximal antidiuresis**. Low doses still achieve **33–50% improvement** in many patients. Supra-threshold plasma levels **prolong antidiuresis**, so low doses also avoid effects on fluid handling beyond the sleep period.

- **Regular sodium monitoring is required.**

COMPARISON — SODIUM MONITORING PROTOCOLS FOR DESMOPRESSIN

Source	Schedule	Higher-risk patients
US FDA prescribing information (DANS, ODT)	Before starting, restarting or dose escalation → within 7 days → ≈30 days → periodically	More frequent if ≥65 years or other risk factors
EAU Male LUTS (strong, LoE 1b)	Baseline → day 3 → day 7 → 1 month → periodically (also before dose titration)	More frequent if ≥65 years or at risk
EAU Male panel on nocturia (grade A, LoE 1a)	Screen at baseline, during titration and during treatment	—
EAU Female LUTS (strong, LoE 1a)	Monitor carefully in the elderly	Avoid if baseline sodium is below normal
ICS	Comprehensive monitoring when warranted; no protocol	—
AUA	No guidance	—
Proposed protocol from pooled phase III ODT data	Baseline sodium in all; stop if any value <135 mmol/L	Repeat at 7 and 30 days if ≥65 years or at risk

- **Independent risk factors for sodium <130 mmol/L** (pooled phase III ODT data): **age ≥65 years; baseline sodium <135 mmol/L; CrCl <60 mL/min; ODT above US-approved doses (>25 mcg women, >50 mcg men).**
- **Most events occur within 3 weeks** of starting ODT (**within 2 weeks** for DANS).
- Modelling the proposed protocol — **less stringent than the FDA's**, with no repeat test in healthy younger patients — **would have prevented every case of sodium <130 mmol/L** at approved ODT doses in both sexes. **Strict monitoring substantially reduces hyponatraemia.**
- **Trap:** FDA and EAU schedules differ — EAU adds a **day-3** check.

COMPARISON — GUIDELINE RECOMMENDATIONS FOR DESMOPRESSIN AND DIURETICS

Guideline	Recommendation
EAU Male LUTS 2022	Offer desmopressin for NP in men <65 years (weak, LoE 1b); offer low-dose desmopressin to men >65 years with ≥2 voids per night due to NP (weak, LoE 1b); discuss benefit versus risk, especially >65 years (strong, LoE 1b)
EAU Male panel on nocturia 2017	Desmopressin may be prescribed in men <65 with hyponatraemia screening (grade A, LoE 1a); trial of timed diuretic for NP , with sodium screening at baseline and during treatment (grade C, LoE 1b)
EAU Female LUTS 2022	Offer desmopressin for NP after counselling, including hyponatraemia (strong , LoE 1b); offer anticholinergic + desmopressin for OAB with NP (weak, LoE 1b); offer timed diuretic for polyuria after counselling (weak, LoE 1b)

- **Pattern: age 65 is the pivot** — standard use under 65, low dose and explicit risk discussion over 65. Only the female guideline makes desmopressin a **strong** recommendation.

Diuretics

- Population studies link **diuretic use and nocturia**. **Furosemide**, the commonest US diuretic, **increases urine output for 6–8 h** (without chronic kidney disease).

- Taken in the hours before sleep, furosemide **causes NP**; taken **>6 h before bed**, it does not raise NUV. So the association is probably **mostly confounding by volume-overload disease** (diuretic use marks the comorbidity) rather than a direct drug effect.
- **Appropriately timed (morning or early-afternoon) diuretics** are a **primary treatment for NP in otherwise healthy adults** — **level 1b** evidence. They work by **restoring the circadian rhythm of natriuresis**, shifting output into the day.

Diuretic late in the day	Diuretic in the morning or early afternoon
Causes NP — diuresis overlaps sleep	Treats NP — natriuresis completed before bed

KEY POINTS: ADDITIONAL TREATMENT CONSIDERATIONS

- **Desmopressin**: the only US-approved nocturia drug — DANS/ODT for **≥2 voids with NP**, lower hyponatraemia risk than old tablets.
- **Women more sensitive** (sex-specific ODT); hyponatraemia is **age- and dose-dependent**, mostly **within the first 2–3 weeks**; **strict sodium monitoring** prevents most.
- **Afternoon diuretics**: large effect in small studies.

Concept — *Refractory nocturia is treated by moving urine production out of the night. Desmopressin suppresses night-time water diuresis; timed diuretics finish the sodium diuresis before bed. Either way the price is sodium balance, so dose, age and monitoring define safe use.*

ANTIDIURETIC AND DIURETIC THERAPY: KEY TRIALS AND PHYSIOLOGIC CONSIDERATIONS

Desmopressin Acetate Nasal Spray

- Approval rested on **two phase III placebo RCTs** (pooled): adults **≥50 years** reporting **≥2** episodes and confirmed **≥2.16** nightly voids over 2-week screening; **1.66 or 0.83 mcg/0.1 mL** (1.11 mcg in one study only) vs placebo, 12 weeks.
- **Co-primary endpoints**: change in frequency; **≥50% responder rate**. Both doses beat placebo on both (see Table 112.7). Sodium checked every 2 weeks.
- **Hyponatraemia** (≤ 125 mmol/L, or <130 with symptoms): **0% at 0.83 mcg, 1.1% at 1.66 mcg, 0.2% placebo**.

Desmopressin Orally Disintegrating Tablets

- **Women are more sensitive to DDAVP** — greater effect on NUV and longer action.
 - Animal studies: **greater renal V2 expression and response in females**.
 - Human physiology: men show **greater osmotic sensitivity of AVP release** but **lower renal sensitivity to AVP**.
 - Possible basis: the **V2 receptor gene is on the X chromosome** (differential expression with X-inactivation); sex hormones may modulate.
- **Consequence**: women do as well on a lower dose and are at **risk of hyponatraemia at higher doses**.

Development path: 4-week dose-ranging trial (10–100 mcg, both sexes) → post hoc: **lower doses effective only in women** → two 12-week sex-specific trials (**25 mcg women; 50 or 75 mcg men**) → Japanese sex-specific trials in NP refractory to behavioural counselling.

- In the dose-ranging trial, **most hyponatraemia occurred in adults ≥65 years on 100 mcg** — above the US maximum dose.
- In men, **75 mcg added little efficacy over 50 mcg but markedly more hyponatraemia** — hence approval of 50 mcg.
- Japanese trials: men improved at 25 and 50 mcg; **women on 25 mcg did not reach significance** on the primary endpoint. However, improvement continued in both sexes in a **long-term extension**, with the **largest effect in women on open-label 25 mcg after 1 year**.

TABLE 112.7 PHASE III CLINICAL TRIALS INVOLVING DESMOPRESSIN TREATMENT IN PATIENTS WITH NOCTURIA

Trial	Population	Dose; duration	Primary result vs placebo	Sodium <130 mmol/L
Oral tablets (earlier, high-dose) — dose-titration, then responders randomised; 3 weeks; primary endpoint 50% responder rate				
Mattiasson 2002	Men ≥20 years, ≥2 voids, Ni >1	100–400 mcg titrated	Response 34% vs 3%	4%, during titration
Lose 2003	Women ≥18 years, ≥2 voids, Ni >1	100–400 mcg titrated	Response 46% vs 7%	6% (or symptomatic), during titration
van Kerrebroeck 2007	Adults ≥18 years, ≥2 voids	100–400 mcg titrated	Response 33% vs 11% ; secondary gains incl. FUSP	3%, during titration
Nasal spray (DANS) — 12 weeks				
Kaminetsky 2018 (pooled, n = 1333)	Adults ≥50 years , ≥2 voids; 43% women	0.83 or 1.66 mcg	Frequency –1.4 / –1.5 vs –1.2 ; ≥50% responders 37.9% / 48.7% vs 30.3% (0.83-mcg P = 0.0159 in table, 0.0227 in text)	≤125 or symptomatic: 0% / 1.1% vs 0.2%
ODT				
Weiss 2012 (n = 757)	Adults, ≥2 voids	10, 25, 50, 100 mcg; 4 weeks	Frequency –1.18 (50 mcg), –1.43 (100 mcg) vs –0.86 ; ≥33% responders 71% vs 47% at 100 mcg; post hoc: women responded at lower doses	3.0% (≤125: 1.1%), all within 1 week
Sand 2013 (n = 261)	Women , ≥2 voids	25 mcg ; 12 weeks	Frequency –1.46 vs –1.24 ; ≥33% responder odds ratio (OR) 1.85	2.2% vs 0%; none ≤125
Weiss 2013 (n = 385)	Men , ≥2 voids	50 or 75 mcg ; 12 weeks	Frequency –1.25 / –1.29 vs –0.88 ; responder OR 1.98 / 2.04 (50-mcg P = 0.0009 in table, 0.009 in text)	1.9% / 7.4% vs 0% (≤125: 1.9% / 3.3%)
Yamaguchi 2020 — men (Japan)	Men ≥20 years, ≥2 voids, NP refractory to behavioural counselling	25 or 50 mcg; 12 weeks	Frequency –0.96 / –1.21 vs –0.76 — positive	0.9% / 1.8%; none ≤125
Yamaguchi 2020 — women (Japan)	Women ≥20 years, as above	25 mcg; 12 weeks	Frequency –1.11 vs –0.95 — not significant ; secondary FUSP, NUV, NPi improved	None

Trial size in the Japanese studies is given as 342 (men) and 190 (women) in the table and 338 and 187 in the text — table versus text denominators, most likely randomised versus analysed populations.

- **Purpose:** traces desmopressin from **high-dose titrated tablets to low-dose, sex- or age-stratified formulations** with less hyponatraemia.
- **Pattern:** the **placebo response is large** (about -0.8 to -1.2 voids), so absolute gains over placebo are **fractions of a void**. Hyponatraemia tracks **dose** ($75 > 50$ mcg; $1.66 > 0.83$ mcg) and appears **early**.
- **Traps:** responder thresholds differ — **50%** (tablets, DANS) vs **33%** (ODT); DANS trials enrolled only **≥ 50 years**; the **only negative primary endpoint is Japanese women on 25 mcg**.

Diuretics

COMPARISON — TIMED DIURETIC TRIALS FOR NOCTURIA

Design	Regimen	Verdict
Placebo-controlled crossover, general-practice adults with ≥ 2 voids	Bumetanide 1 mg , 4–6 h before bed, 2 weeks	Fewer weekly night voids
Placebo RCT, men ≥ 50 years with NP	Furosemide 40 mg , 6 h before bed, 4 weeks	Fewer voids; more responders by ≥ 1 void; no adverse events
RCT, LUTS	Afternoon furosemide 40 mg + doxazosin 4 mg vs doxazosin alone	Combination reduced frequency and NUV
NP patients	Afternoon furosemide 20 mg vs gosha-jinki-gan (East Asian herbal medicine)	Furosemide better for frequency and volume
RCT, older adults	Afternoon furosemide + bedtime DDAVP vs placebo	Improved severity — but loop diuretics are a contraindication to DDAVP (US prescribing information)

- **Pattern:** small trials, consistent benefit; **timing is the active ingredient**.
- **Trap:** combining afternoon furosemide with desmopressin worked in a trial but is **contraindicated** by US labelling.

Concept — The trials turn one physiological fact into dosing rules. **Women are more sensitive to V2 agonism**, so they get lower doses and still risk hyponatraemia at higher ones. The **placebo response is large**, so real gains are modest and must be weighed against sodium risk.

FUTURE DIRECTIONS

- **Patient-centred diagnosis:** user-friendly **electronic bladder diaries**, **remote visits**, comorbidity-driven questionnaires for multidisciplinary care.
- **Single-void nocturnal urine samples** (volume, **osmolality**) collected at the first night void and analysed on standard laboratory equipment — a possible alternative way to **diagnose NP**.
- **Point-of-care biomarkers:** **urinary AVP in early-morning urine** as a potential NP marker, especially with impaired circadian AVP rhythm; **plasma copeptin predicts DDAVP response in children with NP-related enuresis** — possibly relevant to adults.
- **Nocturnal urine composition is heterogeneous:** patients differ in the balance of water and sodium diuresis — the basis for **individualised treatment**.

Predominant nocturnal water diuresis	Predominant nocturnal natriuresis
Likely greater response to DDAVP	Afternoon diuretic likely more relevant
• New agents: DDAVP is the only commercial antidiuretic peptide for nocturia; small-molecule V2 agonists, including truncated AVP analogues with better receptor selectivity and preserved potency, may improve the efficacy-safety balance. • Research need: large prospective trials of behavioural and drug treatments, especially multicomponent interventions targeting nocturnal production.	

Concept — *The field is moving from "nocturnal polyuria" as one diagnosis to **phenotyping what the night urine is made of**, so that the antidiuretic or the natriuretic arm can be matched to the patient.*